

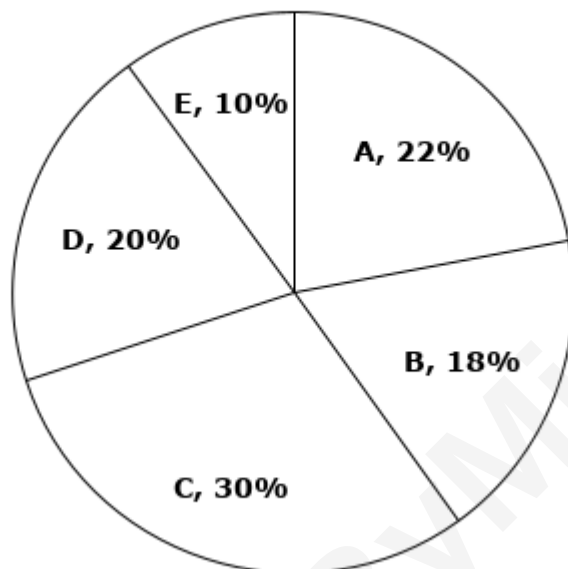
1. Questions

Study the following information carefully and answer the following questions.

The given pie chart shows the percentage distribution of the total number of heavy vehicles in five different cities namely A, B, C, D and E respectively.

Note: The total number of heavy vehicles in all cities together is 12000.

Percentage distribution of the total number of heavy vehicles



The given table chart shows the ratio of the number of trucks to lorries in five different cities namely A, B, C, D and E respectively.

Cities	Ratio of the number of trucks to lorry
A	1:2
B	5:1
C	4:5
D	1:3
E	5:7

In city B, 20% of the trucks are type P, 35% of the trucks are type Q and the rest of them are type R. Find the difference between the sum of the number of type P and R trucks and type Q trucks.

- a. 540
- b. 620
- c. 720

d. 700

e. 510

2. Questions

The total number of heavy vehicles in city F is 30% more than that in city E and the ratio of the number of trucks in city F to E is $x:2$. If the number of lorries in city F is 810, then find the value of x .

a. 5

b. 1

c. 3

d. 7

e. 11

3. Questions

In city C, the number of buses is $\frac{5}{8}$ more than the number of lorries. Find the sum of the number of buses and trucks.

a. 4850

b. 4000

c. 3750

d. 2880

e. 5290

4. Questions

Find the ratio between the number of trucks in cities A and B together to the number of lorries in city E.

a. 56: 81

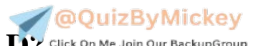
b. 134: 35

c. 121: 99

d. 144: 101

e. 72: 53

5. Questions

The number of trucks in city B is what percentage of the number of lorries in city D?  @QuizByMickey
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a. 120%

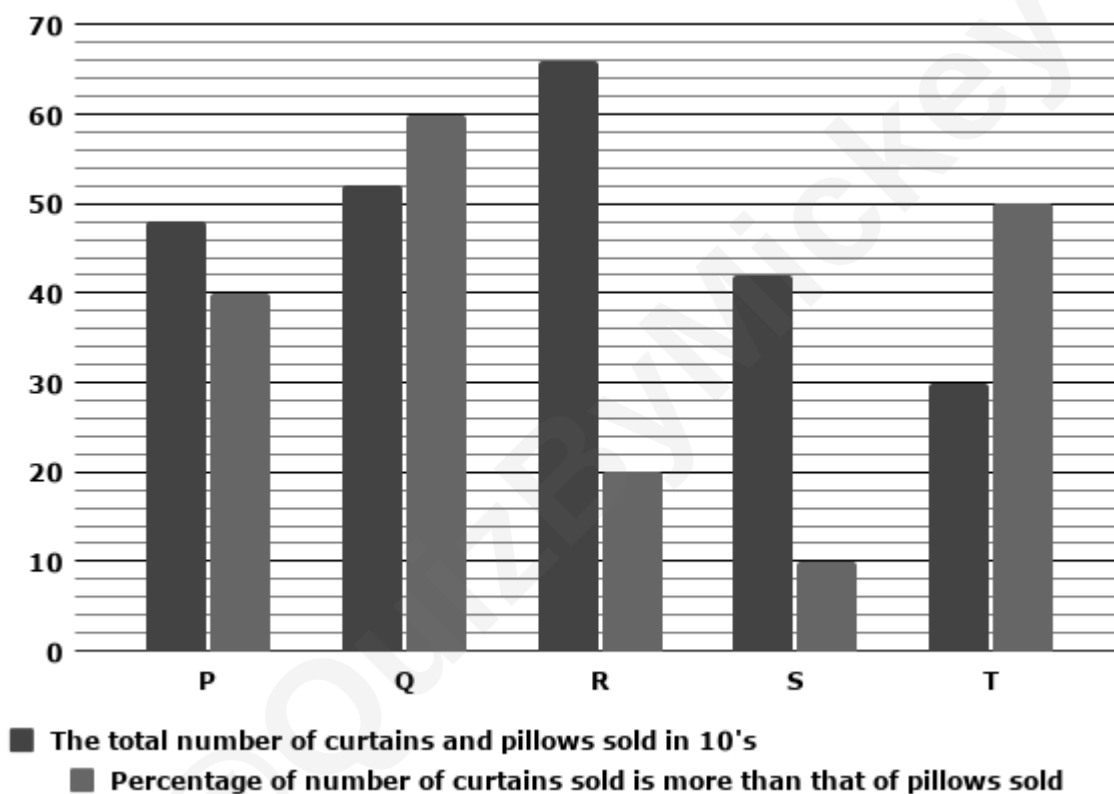
b. 80%

- c. 60%
- d. 100%
- e. 125%

6. Questions

Study the following information carefully and answer the following questions.

The given pie chart shows the total number of curtains and pillows sold (in 10's) in five different shops namely P, Q, R, S and T respectively and also the percentage of how much more the number of curtains sold than the pillows sold in these five different shops.



In shop P, the ratio of the number of green curtains to pillows sold is 8:9 and the ratio of the number of blue curtains to pillows sold is 6:1. Find the number of green curtains sold. (only green and blue colours sold)

- a. 120
- b. 160
- c. 200
- d. 220
- e. 80

7. Questions

The total number of curtains and pillows sold in shop U is $\frac{1}{11}$ less than that in shop R. If the

number of curtains sold in shop U is 20% less than that in shop R, then find the number of pillows sold in shop U.

- a. 250
- b. 300
- c. 312
- d. 425
- e. 400

8. Questions

In shop S, the ratio of the number of curtains sold online to offline is 5:6. If the number of pillows sold offline is 40 less than that of curtains sold, then find the number of pillows sold online.

- a. 125
- b. 80
- c. 60
- d. 100
- e. 120

9. Questions

Find the difference between the total number of curtains and pillows sold in shop Q and the number of pillows sold in shop T.

- a. 350
- b. 320
- c. 400
- d. 450
- e. 310

10. Questions

Find the total number of pillows sold in shop P, R and S together.

- a. 650
- b. 900
- c. 700
- d. 880
- e. 520

11. Questions

Mixture A contains juice and water in the ratio of 7:5. Mixture B contains 25% more juice than A and 20% less water than A, and the total quantity of juice and water in mixture B is 765 litres. Find the initial quantity of water in mixture A.

- a. 300 litres
- b. 420 litres
- c. 450 litres
- d. 320 litres
- e. 250 litres

12. Questions

A and B together can complete a piece of work in 15 days, B alone can complete the same work in 30 days. The ratio of efficiency of A to C is 2:5. If A and B together can complete 80% of the work and the remaining is completed by C alone, then find the total time taken to complete the work.

- a. 16 days
- b. 14.4 days
- c. 22 days
- d. 18.5 days
- e. 11.5 days

13. Questions

Danil invested equal sums in two different schemes, A and B, for 2 years each. Scheme A offers simple interest of 20% p.a., whereas Scheme B offers 10% p.a. compound interest. If the total interest received from both schemes is Rs. 1220, then find the amount invested in A.

- a. Rs. 4000
- b. Rs. 2000
- c. Rs. 2500
- d. Rs. 3000
- e. Rs. 5200

14. Questions

The ratio of the present age of A to B is 6:7. A's age 4 years hence is equal to B's age 2 years ago. If the present age of C is 12 years more than that of A, then find the present age of C.  @QuizByMickey
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- a. 48 years
- b. 42 years

- c. 36 years
- d. 54 years
- e. 28 years

15. Questions

An article is sold for Rs. 2500 after allowing a discount of 20% at a gain of 25%. If the cost price of the article is increased by Rs. 400, then find the difference between the marked price and the cost price of the article.

- a. Rs. 556
- b. Rs. 625
- c. Rs. 715
- d. Rs. 800
- e. Rs. 725

16. Questions

The total surface area of the cube is 864 cm^2 . The length of the cuboid is 2 cm less than the side of the cube, the breadth of the cuboid is half of the side of the cube and the height is 8 cm. Find the volume of the cuboid.

- a. 225 cm^3
- b. 480 cm^3
- c. 420 cm^3
- d. 360 cm^3
- e. 210 cm^3

17. Questions

A car covered a distance of 480 km in 8 hours. The ratio of the speed of the bike to the speed of the car is 4: 3 and the ratio of the speed of the bus to bike is 3: 2. Find the total time taken by bus and car to travel 240 km each .

- a. 8 hours
- b. 11 hours
- c. 6 hours
- d. 12 hours
- e. 15 hours

18. Questions

A and B entered into a partnership by investing Rs. x and Rs. $(x + 1500)$, respectively. After 5 months, A added Rs. $1000/7$, and B withdrew his investment. If the ratio of the annual profit shares of A and B is 26:25, then find the value of x .

- a. 1200
- b. 2000
- c. 3200
- d. 1000
- e. 600

19. Questions

The speed of the boat in still water is 45% more than that of the stream. If the difference between the speed of the boat in still water and the speed of the stream is 18 km/hr, then find the time taken by the boat to travel 319 km in still water.

- a. 3.4 hours
- b. 4 hours
- c. 5.5 hours
- d. 8 hours
- e. 7.5 hours

20. Questions

The total population in 2021 is 36000 and the total population will increase by 20% and 25% respectively in 2022 and 2023 respectively. The ratio of the number of male to females in 2023 is 5:4. Find the number of females in 2023.

- a. 32000
- b. 30000
- c. 21000
- d. 24000
- e. 12000

21. Questions

What value should come in the place of (?) in the following questions.

$$(80 * 5) + 120 - ? = 140 * 5/7$$

- a. 340
- b. 420

- c. 400
- d. 300
- e. 280

22. Questions

$$(\sqrt{225}) * 8 + 16 * 5 = ? - 110$$

- a. 300
- b. 270
- c. 455
- d. 400
- e. 310

23. Questions

$$88\% \text{ of } 600 - 52\% \text{ of } 300 = ? * 2$$

- a. 134
- b. 192
- c. 201
- d. 186
- e. 200

24. Questions

$$^3\sqrt{2197} * \sqrt{361} / \sqrt{169} = ?^2 - 45$$

- a. 8
- b. 12
- c. 9
- d. 4
- e. 3

25. Questions

$$356 - 21 * 4 + 50 = ? - 88$$

- a. 388
- b. 410
- c. 870

d. 720

e. 320

26. Questions

What approximate value should come in the place of (?) in the following questions.

$30.12\% \text{ of } 899.98 + 54.12 = ? - 11.2$

a. 412

b. 330

c. 280

d. 335

e. 488

27. Questions

$(24.21)^2 - (5.98)^2 + 35.22\% \text{ of } 900.12 = ?$

a. 816

b. 855

c. 912

d. 900

e. 456

28. Questions

$44.3 * 1.98 - 6^2 + 12.01 * 3 = ?$

a. 88

b. 90

c. 56

d. 44

e. 23

29. Questions

$9.12\% \text{ of } 1200.12 = 810.12 - ? + 31.12$

a. 815

b. 899

c. 733

- d. 900
- e. 1020

30. Questions

40.12% of $(15.12 * 300/2.89) = ?$

- a. 720
- b. 800
- c. 1260
- d. 600
- e. 900

31. Questions

What value should come in the place of (?) in the following number series?

8, ?, 80, 125, 176, 233

- a. 24
- b. 32
- c. 60
- d. 41
- e. 64

32. Questions

12, ?, 270, 1080, 3780, 11340

- a. 80
- b. 60
- c. 215
- d. 200
- e. 165

33. Questions

48, 75, ?, 264, 480, 823

- a. 145
- b. 155
- c. 139

d. 211

e. 325

34. Questions

?, 18, 40, 84, 172, 348

a. 12

b. 4

c. 2

d. 7

e. 15

35. Questions

5, 15, 60, ?, 1800, 12600

a. 240

b. 350

c. 300

d. 420

e. 1200

36. Questions

Find out the wrong number in the following number series.

55, 68, 85, 104, 159, 156

a. 156

b. 104

c. 85

d. 68

e. 159

37. Questions

15, 40, 89, 172, 291, 460

a. 291

b. 460

c. 40

- d. 89
- e. 172

38. Questions**160, 139, 181, 118, 204, 97**

- a. 204
- b. 97
- c. 181
- d. 139
- e. 160

39. Questions**41, 21, 26, 34.5, 71, 180**

- a. 71
- b. 180
- c. 26
- d. 41
- e. 34.5

40. Questions**214, 206, 190, 169, 134, 94**

- a. 169
- b. 134
- c. 94
- d. 214
- e. 206

41. Questions

The following question contains two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

i). $x^2 - 34x + 288 = 0$

ii). $y^2 - 26y + 168 = 0$

- a. $x > y$

- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

42. Questions

i). $x^2 - 6x - 432 = 0$

ii). $y^2 - 19y + 60 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

43. Questions

i). $x^2 + 21x + 108 = 0$

ii). $y^2 + 25y + 156 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

44. Questions

i). $2x^2 - 15x + 28 = 0$

ii). $y^2 - 8y + 12 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

45. Questions

i). $x^2 - 10x + 21 = 0$

ii). $2y^2 - 13y + 21 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

Explanations:

1. Questions

In city A:

The total number of heavy vehicles = $12000 * 22/100 = 2640$

The number of trucks = $2640 * 1/3 = 880$

The number of lorries = $2640 * 2/3 = 1760$

In city B:

The total number of heavy vehicles = $12000 * 18/100 = 2160$

The number of trucks = $2160 * 5/6 = 1800$

The number of lorries = $2160 * 1/6 = 360$

In city C:

The total number of heavy vehicles = $12000 * 30/100 = 3600$

The number of trucks = $3600 * 4/9 = 1600$

The number of lorries = $3600 * 5/9 = 2000$

In city D:

The total number of heavy vehicles = $12000 * 20/100 = 2400$

The number of trucks = $2400 * 1/4 = 600$

The number of lorries = $2400 * 3/4 = 1800$

In city E:

The total number of heavy vehicles = $12000 * 10/100 = 1200$

The number of trucks = $1200 * 5/12 = 500$

The number of lorries = $1200 * 7/12 = 700$

City	The total number of heavy vehicles	The number of trucks	The number of lorries
A	2640	880	1760
B	2160	1800	360
C	3600	1600	2000
D	2400	600	1800
E	1200	500	700

Answer: A

The number of type P trucks in city B = $1800 * 20/100 = 360$

The number of type Q trucks in city B = $1800 * 35/100 = 630$

The number of type R trucks in city B = $1800 * 45/100 = 810$

Required difference = $(360 + 810) - 630 = (1170 - 630) = 540$

Hence, the correct answer is option A

2. Questions

In city A:

The total number of heavy vehicles = $12000 * 22/100 = 2640$

The number of trucks = $2640 * 1/3 = 880$

The number of lorries = $2640 * 2/3 = 1760$

In city B:

The total number of heavy vehicles = $12000 * 18/100 = 2160$

The number of trucks = $2160 * 5/6 = 1800$

The number of lorries = $2160 * 1/6 = 360$

In city C:

The total number of heavy vehicles = $12000 * 30/100 = 3600$

The number of trucks = $3600 * 4/9 = 1600$

The number of lorries = $3600 * 5/9 = 2000$

In city D:

The total number of heavy vehicles = $12000 * 20/100 = 2400$

The number of trucks = $2400 * \frac{1}{4} = 600$

The number of lorries = $2400 * \frac{3}{4} = 1800$

In city E:

The total number of heavy vehicles = $12000 * \frac{10}{100} = 1200$

The number of trucks = $1200 * \frac{5}{12} = 500$

The number of lorries = $1200 * \frac{7}{12} = 700$

City	The total number of heavy vehicles	The number of trucks	The number of lorries
A	2640	880	1760
B	2160	1800	360
C	3600	1600	2000
D	2400	600	1800
E	1200	500	700

Answer: C

The total number of heavy vehicles in city F = $1200 * \frac{130}{100} = 1560$

The number of lorries in city F = 810

The number of trucks in city F = $(1560 - 810) = 750$

$x:2 = 750:500$

$x:2 = 3:2$

$x = 3$

Hence, the correct answer is option C

3. Questions

In city A:

The total number of heavy vehicles = $12000 * \frac{22}{100} = 2640$

The number of trucks = $2640 * \frac{1}{3} = 880$

The number of lorries = $2640 * \frac{2}{3} = 1760$

In city B:

The total number of heavy vehicles = $12000 * \frac{18}{100} = 2160$

The number of trucks = $2160 * \frac{5}{6} = 1800$

The number of lorries = $2160 * \frac{1}{6} = 360$

In city C:

The total number of heavy vehicles = $12000 * \frac{30}{100} = 3600$

The number of trucks = $3600 * \frac{4}{9} = 1600$

The number of lorries = $3600 * \frac{5}{9} = 2000$

In city D:

The total number of heavy vehicles = $12000 * \frac{20}{100} = 2400$

The number of trucks = $2400 * \frac{1}{4} = 600$

The number of lorries = $2400 * \frac{3}{4} = 1800$

In city E:

The total number of heavy vehicles = $12000 * \frac{10}{100} = 1200$

The number of trucks = $1200 * \frac{5}{12} = 500$

The number of lorries = $1200 * \frac{7}{12} = 700$

City	The total number of heavy vehicles	The number of trucks	The number of lorries
A	2640	880	1760
B	2160	1800	360
C	3600	1600	2000
D	2400	600	1800
E	1200	500	700

Answer: A

The number of buses in city C = $2000 + 2000 * \frac{5}{8} = (2000 + 1250) = 3250$

The number of trucks in city C = 1600

Required sum = $(3250 + 1600) = 4850$

Hence, the correct answer is option A

4. Questions

In city A:

The total number of heavy vehicles = $12000 * \frac{22}{100} = 2640$

The number of trucks = $2640 * \frac{1}{3} = 880$

The number of lorries = $2640 * \frac{2}{3} = 1760$

In city B:

The total number of heavy vehicles = $12000 * \frac{18}{100} = 2160$

The number of trucks = $2160 * \frac{5}{6} = 1800$

The number of lorries = $2160 * \frac{1}{6} = 360$

In city C:

The total number of heavy vehicles = $12000 * \frac{30}{100} = 3600$

The number of trucks = $3600 * \frac{4}{9} = 1600$

The number of lorries = $3600 * \frac{5}{9} = 2000$

In city D:

The total number of heavy vehicles = $12000 * \frac{20}{100} = 2400$

The number of trucks = $2400 * \frac{1}{4} = 600$

The number of lorries = $2400 * \frac{3}{4} = 1800$

In city E:

The total number of heavy vehicles = $12000 * \frac{10}{100} = 1200$

The number of trucks = $1200 * \frac{5}{12} = 500$

The number of lorries = $1200 * \frac{7}{12} = 700$

City	The total number of heavy vehicles	The number of trucks	The number of lorries
A	2640	880	1760
B	2160	1800	360
C	3600	1600	2000
D	2400	600	1800
E	1200	500	700

Answer: B

The number of trucks in cities A and B together = $(880 + 1800) = 2680$

The number of lorries in city E = 700

Required ratio = $2680: 700 = 134: 35$

Hence, the correct answer is option B

5. Questions

In city A:

The total number of heavy vehicles = $12000 * 22/100 = 2640$

The number of trucks = $2640 * 1/3 = 880$

The number of lorries = $2640 * 2/3 = 1760$

In city B:

The total number of heavy vehicles = $12000 * 18/100 = 2160$

The number of trucks = $2160 * 5/6 = 1800$

The number of lorries = $2160 * 1/6 = 360$

In city C:

The total number of heavy vehicles = $12000 * 30/100 = 3600$

The number of trucks = $3600 * 4/9 = 1600$

The number of lorries = $3600 * 5/9 = 2000$

In city D:

The total number of heavy vehicles = $12000 * 20/100 = 2400$

The number of trucks = $2400 * 1/4 = 600$

The number of lorries = $2400 * 3/4 = 1800$

In city E:

The total number of heavy vehicles = $12000 * 10/100 = 1200$

The number of trucks = $1200 * 5/12 = 500$

The number of lorries = $1200 * 7/12 = 700$

City	The total number of heavy vehicles	The number of trucks	The number of lorries
A	2640	880	1760
B	2160	1800	360
C	3600	1600	2000
D	2400	600	1800
E	1200	500	700

Answer: D

The number of trucks in city B = 1800

The number of lorries in city D = 1800

Required percentage = $(1800/1800) * 100 = 100\%$

Hence, the correct answer is option D

6. Questions

In shop P:

The total number of curtains and pillows sold = 480

The ratio of the number of curtains to pillows sold = 140: 100 = 7:5

The number of curtains sold = $480 * 7/12 = 280$

The number of pillows sold = $480 * 5/12 = 200$

In shop Q:

The total number of curtains and pillows sold = 520

The ratio of the number of curtains to pillows sold = 160: 100 = 8: 5

The number of curtains sold = $520 * 8/13 = 320$

The number of pillows sold = $520 - 320 = 200$

In shop R:

The total number of curtains and pillows sold = 660

The ratio of the number of curtains to pillows sold = 120: 100 = 6: 5

The number of curtains sold = $660 * 6/11 = 360$

The number of pillows sold = $660 * 5/11 = 300$

In shop S:

The total number of curtains and pillows sold = 420

The ratio of the number of curtains to pillows sold = 110: 100 = 11: 10

The number of curtains sold = $420 * 11/21 = 220$

The number of pillows sold = $420 * 10/21 = 200$

In shop T:

The total number of curtains and pillows sold = 300

The ratio of the number of curtains to pillows sold = 150: 100 = 3:2

The number of curtains sold = $300 * 3/5 = 180$

The number of pillows sold = $300 * 2/5 = 120$

Shop	The total number of curtains and pillows sold	The number of curtains sold	The number of pillows sold
P	480	280	200
Q	520	320	200
R	660	360	300
S	420	220	200
T	300	180	120

Answer: B

Let, the number of green curtains and pillows sold in shop P be $8x$ and $9x$ respectively.

Let, the number of blue curtains and pillows sold in shop P be $6y$ and y respectively.

$$8x + 6y = 280 \text{ ----> (1)}$$

$$9x + y = 200 \text{ -----> (2)}$$

Multiplying equation (2) * 6, we get

$$54x + 6y = 1200 \text{ ----> (3)}$$

By solving equations (1) and (3), we get

$$x = 20 \text{ and } y = 20$$

The number of green curtains sold in shop P = $8 * 20 = 160$

Hence, the correct answer is option B

7. Questions

In shop P:

The total number of curtains and pillows sold = 480

The ratio of the number of curtains to pillows sold = $140:100 = 7:5$

The number of curtains sold = $480 * 7/12 = 280$

The number of pillows sold = $480 * 5/12 = 200$

In shop Q:

The total number of curtains and pillows sold = 520

The ratio of the number of curtains to pillows sold = $160:100 = 8:5$

The number of curtains sold = $520 * 8/13 = 320$

The number of pillows sold = $520 - 320 = 200$

In shop R:

The total number of curtains and pillows sold = 660

The ratio of the number of curtains to pillows sold = $120:100 = 6:5$

The number of curtains sold = $660 * 6/11 = 360$

The number of pillows sold = $660 * 5/11 = 300$

In shop S:

The total number of curtains and pillows sold = 420

The ratio of the number of curtains to pillows sold = $110:100 = 11:10$

The number of curtains sold = $420 * 11/21 = 220$

The number of pillows sold = $420 * 10/21 = 200$

In shop T:

The total number of curtains and pillows sold = 300

The ratio of the number of curtains to pillows sold = $150:100 = 3:2$

The number of curtains sold = $300 * 3/5 = 180$

The number of pillows sold = $300 * 2/5 = 120$

Shop	The total number of curtains and pillows sold	The number of curtains sold	The number of pillows sold
P	480	280	200
Q	520	320	200
R	660	360	300
S	420	220	200
T	300	180	120

Answer: C

The total number of curtains and pillows sold in shop U = $660 - 660/11 = (660 - 60) = 600$

The number of curtains sold in shop U = $360 * 80/100 = 288$

The number of pillows sold in shop U = $(600 - 288) = 312$

Hence, the correct answer is option C

8. Questions

In shop P:

The total number of curtains and pillows sold = 480

The ratio of the number of curtains to pillows sold = 140: 100 = 7:5

The number of curtains sold = $480 \times \frac{7}{12} = 280$

The number of pillows sold = $480 \times \frac{5}{12} = 200$

In shop Q:

The total number of curtains and pillows sold = 520

The ratio of the number of curtains to pillows sold = 160: 100 = 8: 5

The number of curtains sold = $520 \times \frac{8}{13} = 320$

The number of pillows sold = $520 - 320 = 200$

In shop R:

The total number of curtains and pillows sold = 660

The ratio of the number of curtains to pillows sold = 120: 100 = 6: 5

The number of curtains sold = $660 \times \frac{6}{11} = 360$

The number of pillows sold = $660 \times \frac{5}{11} = 300$

In shop S:

The total number of curtains and pillows sold = 420

The ratio of the number of curtains to pillows sold = 110: 100 = 11: 10

The number of curtains sold = $420 \times \frac{11}{21} = 220$

The number of pillows sold = $420 \times \frac{10}{21} = 200$

In shop T:

The total number of curtains and pillows sold = 300

The ratio of the number of curtains to pillows sold = 150: 100 = 3:2

The number of curtains sold = $300 \times \frac{3}{5} = 180$

The number of pillows sold = $300 \times \frac{2}{5} = 120$

Shop	The total number of curtains and pillows sold	The number of curtains sold	The number of pillows sold
P	480	280	200
Q	520	320	200
R	660	360	300
S	420	220	200
T	300	180	120

Answer: E

The number of curtains sold offline in shop S = $220 \times \frac{6}{11} = 120$

The number of pillows sold offline in shop S = $120 - 40 = 80$

The number of pillows sold online in shop S = $200 - 80 = 120$

Hence, the correct answer is option E

9. Questions

In shop P:

The total number of curtains and pillows sold = 480

The ratio of the number of curtains to pillows sold = 140: 100 = 7:5

The number of curtains sold = $480 \times \frac{7}{12} = 280$

The number of pillows sold = $480 \times \frac{5}{12} = 200$

In shop Q:

The total number of curtains and pillows sold = 520

The ratio of the number of curtains to pillows sold = 160: 100 = 8: 5

The number of curtains sold = $520 \times \frac{8}{13} = 320$

The number of pillows sold = $520 - 320 = 200$

In shop R:

The total number of curtains and pillows sold = 660

The ratio of the number of curtains to pillows sold = 120: 100 = 6: 5

The number of curtains sold = $660 \times \frac{6}{11} = 360$

The number of pillows sold = $660 \times \frac{5}{11} = 300$

In shop S:

The total number of curtains and pillows sold = 420

The ratio of the number of curtains to pillows sold = 110: 100 = 11: 10

The number of curtains sold = $420 \times \frac{11}{21} = 220$

The number of pillows sold = $420 \times \frac{10}{21} = 200$

In shop T:

The total number of curtains and pillows sold = 300

The ratio of the number of curtains to pillows sold = 150: 100 = 3:2

The number of curtains sold = $300 \times \frac{3}{5} = 180$

The number of pillows sold = $300 \times \frac{2}{5} = 120$

Shop	The total number of curtains and pillows sold	The number of curtains sold	The number of pillows sold
P	480	280	200
Q	520	320	200
R	660	360	300
S	420	220	200
T	300	180	120

Answer: C

The total number of curtains and pillows sold in shop Q = 520

The number of pillows sold in shop T = 120

Required difference = $(520 - 120) = 400$

Hence, the correct answer is option C

10. Questions

In shop P:

The total number of curtains and pillows sold = 480

The ratio of the number of curtains to pillows sold = 140: 100 = 7:5

The number of curtains sold = $480 \times \frac{7}{12} = 280$

The number of pillows sold = $480 \times \frac{5}{12} = 200$

In shop Q:

The total number of curtains and pillows sold = 520

The ratio of the number of curtains to pillows sold = 160: 100 = 8: 5

The number of curtains sold = $520 \times \frac{8}{13} = 320$

The number of pillows sold = $520 - 320 = 200$

In shop R:

The total number of curtains and pillows sold = 660

The ratio of the number of curtains to pillows sold = 120: 100 = 6: 5

The number of curtains sold = $660 \times \frac{6}{11} = 360$

The number of pillows sold = $660 \times \frac{5}{11} = 300$

In shop S:

The total number of curtains and pillows sold = 420

The ratio of the number of curtains to pillows sold = 110: 100 = 11: 10

The number of curtains sold = $420 \times \frac{11}{21} = 220$

The number of pillows sold = $420 \times \frac{10}{21} = 200$

In shop T:

The total number of curtains and pillows sold = 300

The ratio of the number of curtains to pillows sold = 150: 100 = 3:2

The number of curtains sold = $300 \times \frac{3}{5} = 180$

The number of pillows sold = $300 \times \frac{2}{5} = 120$

Shop	The total number of curtains and pillows sold	The number of curtains sold	The number of pillows sold
P	480	280	200
Q	520	320	200
R	660	360	300
S	420	220	200
T	300	180	120

Answer: C

The total number of pillows sold in shop P, R and S together = $(200 + 300 + 200) = 700$

Hence, the correct answer is option C

11. Questions

Answer: A

Let, the initial quantity of juice and water in the mixture A be $7x$ litres and $5x$ litres respectively.

The initial quantity of juice in the mixture B = $7x * 125/100 = 8.75x$

The initial quantity of water in the mixture B = $5x * 80/100 = 4x$ litres

$$8.75x + 4x = 765$$

$$12.75x = 765$$

$$x = 60$$

The initial quantity of water in mixture A = $5 * 60 = 300$ litres

Hence, the correct answer is option A

12. Questions

Answer: B

According to the question,

Let, the total work = 60 units

The combined efficiency of A and B = $60/15 = 4$ units

The efficiency of B = $60/30 = 2$ units

The efficiency of A = $4 - 2 = 2$ units

The efficiency of C = $2 * 5/2 = 5$ units

The 80% of the work completed by A and B = $(60 * 80/100)/4 = 12$ days

The remaining work = $(60 - 48) = 12$ units

The time taken by C to complete the work = $12/5 = 2.4$ days

Required time taken = $(12 + 2.4) = 14.4$ days

Hence, the correct answer is option B

13. Questions

Answer: B

Let, the sum invested = Rs. x

From scheme A,

$$SI = PNR/100$$

$$SI = x * 20/100 * 2$$

$$SI = \text{Rs. } 0.4x$$

From scheme B,

$$CI = P * (1 + R/100)^n - P$$

$$CI = x * 1.1 * 1.1 - x$$

$$CI = \text{Rs. } 0.21x$$

$$0.4x + 0.21x = 1220$$

$$0.61x = 1220$$

$$x = 2000$$

Hence, the correct answer is option B

14. Questions

Answer: A

Let, the present age of A and B be $6x$ years and $7x$ years respectively.

$$(6x + 4) = (7x - 2)$$

$$6x - 7x = -2 - 4$$

$$-x = -6$$

$$x = 6$$

The present age of A = $6 * 6 = 36$ years

The present age of C = $36 + 12 = 48$ years

Hence, the correct answer is option A

15. Questions

Answer: E

The selling price of the article = Rs. 2500

Let, the cost price of the article = Rs. $100x$

$$100x * 125/100 = 2500$$

$$125x = 2500$$

$$x = 20$$

The cost price of the article = $100 * 20 = \text{Rs. } 2000$

Let, the marked price of the article = Rs. y

$$2500 = y * 80/100$$

$$y = 3125$$

The cost price of the article after increased = $(2000 + 400) = \text{Rs. } 2400$

Required difference = $(3125 - 2400) = \text{Rs. } 725$

Hence, the correct answer is option E

16. Questions

Answer: B

The total surface area of the cube = $6a^2 \text{ cm}^2$

$$864 = 6a^2$$

$$144 = a^2$$

$$a = 12 \text{ cm}$$

The length of the cuboid = $12 - 2 = 10 \text{ cm}$

The breadth of the cuboid = $12/2 = 6 \text{ cm}$

The height of the cuboid = 8 cm

The volume of the cuboid = $l * b * h \text{ cm}^3$

$$= 10 * 6 * 8$$

$$= 480 \text{ cm}^3$$

Hence, the correct answer is option B

17. Questions

Answer: C

Distance = Speed * Time

The speed of car = $480/8 = 60 \text{ kmph}$

The speed of the bike = $4/3 * 60 = 80 \text{ kmph}$

The speed of the bus = $3/2 * 80 = 120 \text{ kmph}$

Required time taken = $(240/120) + (240/60) = (2 + 4) = 6 \text{ hours}$

Hence, the correct answer is option C

18. Questions

Answer: D

The ratio of the profit share of A to B = $(x * 5 + (x + 1000/7) * 7) : (x + 1500) * 5$

$$= (5x + 7x + 1000) : (5x + 7500)$$

$$= (12x + 1000) : (5x + 7500)$$

$$(12x + 1000)/(5x + 7500) = 26/25$$

$$300x + 25000 = 130x + 195000$$

$$170x = 170000$$

$$x = 1000$$

Hence, the correct answer is option D

19. Questions

Answer: C

The ratio of the speed of the boat in still water and the speed of the stream = $145:100 = 29:20$

Let, the speed of the boat in still water and the speed of the stream be 29 km/hr and 20 km/hr respectively.

$$29x - 20x = 18$$

$$9x = 18$$

$$x = 2$$

The speed of the boat in still water = $29 * 2 = 58$ km/hr

The speed of the stream = $20 * 2 = 40$ km/hr

Required time taken = $319/58 = 5.5$ hours

Hence, the correct answer is option C

20. Questions

Answer: D

The total population in 2021 = 36000

The total population in 2023 = $36000 * 120/100 * 125/100$
 $= 54000$

The number of females in 2023 = $54000 * 4/9 = 24000$

Hence, the correct answer is option D

21. Questions

Answer: B

$$(80 * 5) + 120 - ? = 140 * 5/7$$

$$400 + 120 - ? = 20 * 5$$

$$520 - ? = 100$$

$$? = 420$$

Hence, the correct answer is option B

22. Questions

Answer: E

$$(\sqrt{225}) * 8 + 16 * 5 = ? - 110$$

$$15 * 8 + 80 + 110 = ?$$

$$120 + 190 = ?$$

$$? = 310$$

Hence, the correct answer is option E

23. Questions

Answer: D

$$88\% \text{ of } 600 - 52\% \text{ of } 300 = ? * 2$$

$$528 - 156 = ? * 2$$

$$? = 186$$

Hence, the correct answer is option D

24. Questions

Answer: A

$$\sqrt[3]{2197} * \sqrt{361} / \sqrt{169} = ?^2 - 45$$

$$13 * 19/13 = ?^2 - 45$$

$$19 + 45 = ?^2$$

$$?^2 = 64$$

$$? = 8$$

Hence, the correct answer is option A

25. Questions

Answer: B

$$356 - 21 * 4 + 50 = ? - 88$$

$$356 - 84 + 50 + 88 = ?$$

$$? = 410$$

Hence, the correct answer is option B

26. Questions

Answer: D

$$30.12\% \text{ of } 899.98 + 54.12 = ? - 11.2$$

$$30\% \text{ of } 900 + 54 = ? - 11$$

$$270 + 54 + 11 = ?$$

$$? = 335$$

Hence, the correct answer is option D

27. Questions

Answer: B

$$(24.21)^2 - (5.98)^2 + 35.22\% \text{ of } 900.12 = ?$$

$$24^2 - 6^2 + 35\% \text{ of } 900 = ?$$

$$576 - 36 + 315 = ?$$

$$? = 855$$

Hence, the correct answer is option B

28. Questions

Answer: A

$$44.3 * 1.98 - 6^2 + 12.01 * 3 = ?$$

$$44 * 2 - 36 + 12 * 3 = ?$$

$$88 - 36 + 36 = ?$$

$$? = 88$$

Hence, the correct answer is option A

29. Questions

Answer: C

$$9.12\% \text{ of } 1200.12 = 810.12 - ? + 31.12$$

$$9\% \text{ of } 1200 = 810 - ? + 31$$

$$108 = 810 - ? + 31$$

$$-733 = - ?$$

$$? = 733$$

Hence, the correct answer is option C

30. Questions

Answer: D

$$40.12\% \text{ of } (15.12 * 300/2.89) = ?$$

$$40\% \text{ of } (15 * 300/3) = ?$$

$$40\% \text{ of } 1500 = ?$$

$$? = 600$$

Hence, the correct answer is option D

31. Questions**Answer: D**

$$8 + (3 * 11) = 41$$

$$41 + (3 * 13) = 80$$

$$80 + (3 * 15) = 125$$

$$125 + (3 * 17) = 176$$

$$176 + (3 * 19) = 233$$

Hence, the correct answer is option D

32. Questions**Answer: B**

$$12 * 5 = 60$$

$$60 * 4.5 = 270$$

$$270 * 4 = 1080$$

$$1080 * 3.5 = 3780$$

$$3780 * 3 = 11340$$

Hence, the correct answer is option B

33. Questions**Answer: C**

$$48 + 3^3 = 75$$

$$75 + 4^3 = 139$$

$$139 + 5^3 = 264$$

$$264 + 6^3 = 480$$

$$480 + 7^3 = 823$$

Hence, the correct answer is option C

34. Questions**Answer: D**

$$7 * 2 + 4 = 18$$

$$18 * 2 + 4 = 40$$

$$40 * 2 + 4 = 84$$

$$84 * 2 + 4 = 172$$

$$172 * 2 + 4 = 348$$

Hence, the correct answer is option D

35. Questions

Answer: C

$$5 * 3 = 15$$

$$15 * 4 = 60$$

$$60 * 5 = \mathbf{300}$$

$$300 * 6 = 1800$$

$$1800 * 7 = 12600$$

Hence, the correct answer is option C

36. Questions

Answer: E

$$55 + 13 = 68$$

$$68 + 17 = 85$$

$$85 + 19 = 104$$

$$104 + 23 = \mathbf{127}$$

$$127 + 29 = 156$$

Hence, the correct answer is option E

37. Questions

Answer: E

$$15 + 5^2 = 40$$

$$40 + 7^2 = 89$$

$$89 + 9^2 = \mathbf{170}$$

$$170 + 11^2 = 291$$

$$291 + 13^2 = 460$$

Hence, the correct answer is option E

38. Questions

Answer: A

$$160 - 21 = 139$$

$$139 + 42 = 181$$

$$181 - 63 = 118$$

$$118 + 84 = \mathbf{202}$$

$$202 - 105 = 97$$

Hence, the correct answer is option A

39. Questions

Answer: C

$$41 * 0.5 + 0.5 = 21$$

$$21 * 1 + 1 = \mathbf{22}$$

$$22 * 1.5 + 1.5 = 34.5$$

$$34.5 * 2 + 2 = 71$$

$$71 * 2.5 + 2.5 = 180$$

Hence, the correct answer is option C

40. Questions

Answer: A

$$214 - 8 = 206$$

$$206 - 16 = 190$$

$$190 - 24 = \mathbf{166}$$

$$166 - 32 = 134$$

$$134 - 40 = 94$$

Hence, the correct answer is option A

41. Questions

Answer: A

i). $x^2 - 34x + 288 = 0$

$$x^2 - 18x - 16x + 288 = 0$$

$$x(x - 18) - 16(x - 18) = 0$$

$$(x - 18)(x - 16) = 0$$

$$x = 18, 16$$

ii). $y^2 - 26y + 168 = 0$

$$y^2 - 14y - 12y + 168 = 0$$

$$y(y - 14) - 12(y - 14) = 0$$

$$(y - 14)(y - 12) = 0$$

$$y = 14, 12$$

Thus, $x > y$

Hence, the correct answer is option A

42. Questions

Answer: C

i). $x^2 - 6x - 432 = 0$

$$x^2 - 24x + 18x - 432 = 0$$

$$x(x - 24) + 18(x - 24) = 0$$

$$(x - 24)(x + 18) = 0$$

$$x = 24, -18$$

ii). $y^2 - 19y + 60 = 0$

$$y^2 - 15y - 4y + 60 = 0$$

$$y(y - 15) - 4(y - 15) = 0$$

$$(y - 15)(y - 4) = 0$$

$$y = 15, 4$$

Thus, the relationship between x and y cannot be established

Hence, the correct answer is option C

43. Questions

Answer: B

i). $x^2 + 21x + 108 = 0$

$$x^2 + 12x + 9x + 108 = 0$$

$$x(x + 12) + 9(x + 12) = 0$$

$$(x + 12)(x + 9) = 0$$

$$x = -12, -9$$

ii). $y^2 + 25y + 156 = 0$

$$y^2 + 12y + 13y + 156 = 0$$

$$y(y + 12) + 13(y + 12) = 0$$

$$(y + 12)(y + 13) = 0$$

$$y = -12, -13$$

Thus, $x \geq y$

Hence, the correct answer is option B

44. Questions

Answer: C

i). $2x^2 - 15x + 28 = 0$

$$2x^2 - 7x - 8x + 28 = 0$$

$$x(2x - 7) - 4(2x - 7) = 0$$

$$(2x - 7)(x - 4) = 0$$

$$x = 3.5, 4$$

ii). $y^2 - 8y + 12 = 0$

$$y^2 - 6y - 2y + 12 = 0$$

$$y(y - 6) - 2(y - 6) = 0$$

$$(y - 6)(y - 2) = 0$$

$$y = 6, 2$$

Thus, the relationship between x and y cannot be established

Hence, the correct answer is option C

45. Questions

Answer: C

i). $x^2 - 10x + 21 = 0$

$$x^2 - 7x - 3x + 21 = 0$$

$$x(x - 7) - 3(x - 7) = 0$$

$$(x - 7)(x - 3) = 0$$

$$x = 7, 3$$

ii). $2y^2 - 13y + 21 = 0$

$$2y^2 - 6y - 7y + 21 = 0$$

$$2y(y - 3) - 7(y - 3) = 0$$

$$(y - 3)(2y - 7) = 0$$

$$y = 3, 3.5$$

Thus, $x = y$ or relationship cannot be established.

Hence, the correct answer is option C

@QuizByMickey